Lecture 5 // Sept 8 // Yang Mills Functional & Equations

Last time: We defined the following objects:

(M^4, g), E an $su(2)$ bundle over M

$\mathcal{A}$:= space of $su(2)$ connections $\{\nabla = d + A\}$ this is an affine space over $T(T^*M \otimes \mathrm{ad}E)$

$\mathcal{G}$ = $SU(2)$ gauge gp $= \Gamma(\mathrm{Aut}E)$

$g \curvearrowright A \quad A \rightsquigarrow g^{-1}Ag + g^{-1}dg$

$\mathcal{M}$ = Moduli space of Yang Mills connections with self dual curvature $= \left\{ \begin{array}{l} SU(2) \text{ connections } \nabla = d + A \\ \text{such that } F_A = *F_A \\ \text{(self dual)} \end{array} \right\} / \mathcal{G}$

This condition of having self-dual curvature is actually a specific instance of a larger phenomena, which we'll discuss today to sit ASD connections in context.

The Yang Mills Functional

Recall that the Lie algebra of $SU(2)$, $su(2)$ is defined as the subset of 2×2 complex matrices which are trace-free and skew-hermitian.

$$su(2) = \left\{ \begin{bmatrix} ix & -y+iz \\ y+iz & -ix \end{bmatrix} \;\middle|\; (x,y,z) \in \mathbb{R}^3 \right\}$$

In particular, this definition clearly shows that $su(2)$ is in bijection with $\mathbb{R}^3$. $su(2)$ can be endowed with an inner product given by $\langle A, B \rangle = \mathrm{tr}(AB^+) = -\mathrm{tr}(AB)$, which is the same inner product as the one on $\mathbb{R}^3$ (the dot product).

$$\left\langle \begin{bmatrix} ix & -y+iz \\ y+iz & -ix \end{bmatrix}, \begin{bmatrix} ip & -q+ir \\ q+ir & -ip \end{bmatrix} \right\rangle = 2(xp+yq+zr) = 2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

This inner product on $\mathfrak{su}(2)$ can be extended to an L^2 inner product on $\Omega^i(\mathrm{ad}E) = \Gamma(\Lambda^i T^*M \otimes \mathrm{ad}E)$.

Given two sections F and G, their inner product is defined as

$$\langle F, G\rangle_{L^2} = -\int_M \mathrm{tr}(F \wedge * G)$$

where the inner product and trace is taken pointwise for each $p \in M$, giving a 4-form at each point, ie, a function of p which we can integrate over M. By taking $G=F$, the norm of F is thus $\|F\|^2 = \langle F, F\rangle_{L^2}$.

Using this inner product, we may now define the Yang-Mills Functional.

Def [Yang-Mills Functional]

$$YM: \{\mathfrak{su}(2)\text{-connections}\} \to \mathbb{R}$$

$$\nabla \mapsto YM(\nabla) = \langle F_A, F_A\rangle_{L^2} = -\int_M \mathrm{tr}(F_A \wedge * F_A) = -\int_M \|F_A\|^2 \mathrm{dvol}_M$$

Heuristically, $YM(\nabla)$ measures the "strength" of curvature of ∇.

For some connection, what does it mean to be a critical point of the Yang Mills functional? Recall that the space of connections is affine over $\Gamma(T^*M \otimes \mathrm{ad}E)$. A fixed connection ∇ is a critical point of YM if every nearby connection ∇' has larger or smaller curvature. Any such ∇' can be written as $\nabla' = \nabla + tB$, ie, a small perturbation of ∇ by some section $B \in \Gamma(T^*M \otimes \mathrm{ad}E)$. In terms of F_A, the curvature of ∇' becomes

$$\begin{aligned} F_{A+tB} &= d(A+tB) + (A+tB)\wedge(A+tB) \\ &= F_A + t(dB + A\wedge B + B\wedge A) + t^2 B\wedge B \end{aligned}$$

Define $D_A B := dB + A\wedge B + B\wedge A$.

The derivative $\frac{d}{dt}\Big|_{t=0} YM(\nabla + tB)$ then becomes

$$\frac{d}{dt}\Big|_{t=0} YM(\nabla + tB) = \frac{d}{dt}\Big|_{t=0} \langle F_{A+tB},\ F_{A+tB} \rangle$$

$$= \frac{d}{dt}\Big|_{t=0} \langle F_A + tD_AB + t^2 B\wedge B,\ F_A + tD_AB + t^2 B\wedge B \rangle$$

$$\vdots$$

$$= 2\ \langle D_AB,\ F_A \rangle$$

Therefore ∇ is a critical point of YM IFF this pairing is 0 for all B.

Before we continue, we should discuss the appearance of D_AB.

Interlude: More about connections & bundles

Recall the usual deRham complex.

$$\Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \cdots$$

Now, take a vector bundle $E \to B$ and a fixed connection ∇ on E

We have already described ∇ as a map $\nabla: \Gamma(E) \to \Gamma(T^*M \otimes E)$

Can we get an analogous complex to the deRham complex?

To do so, we must generalize the spaces & the maps.

Recall that the Ω^i in the deRham complex are defined to be Sections of the bundle $\Lambda^i T^*E \to M$.

Def E-valued i-forms are Sections of the bundle $\Lambda^i T^*M \otimes E$

$$\Omega^i(E) = \Gamma(\Lambda^i T^*M \otimes E)$$

Note that when $i=0$, we recover $\Omega^0(E) = \Gamma(\Lambda^0 T^*M \otimes E) = \Gamma(E)$ and when $i=1$, we get $\Omega^1(E) = \Gamma(\Lambda^1 T^*M \otimes E)$. So as defined above, a connection gives a map $\nabla: \Omega^0(E) \to \Omega^1(E)$. We extend the definition of ∇ using the Leibniz rule, as follows.

Def $\nabla : \Omega^i(E) \to \Omega^{i+1}(E)$

$$\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^i \alpha \wedge \nabla) s$$

where α is an i-form and s = section of E, so $(\alpha \otimes s) \in \Omega^i(E)$

Note that unlike in the deRham complex, $\nabla^2 \neq 0$. However, measuring the failure of this fact is very useful.

Lemma $\nabla^2(s) = F_A s$

Pf Choose basis sections $\{e_1, \ldots, e_n\}$ by choosing a trivialization of the bundle.

Then $s = \sum s_i e_i$, where s_i are functions.

Recall $\nabla = d + A$. Therefore

$$\nabla^2 s = \nabla(\nabla s) = \nabla\left(\sum_{ij} \underbrace{ds_i}_{\text{derivatives of coefcs}} \otimes e_i + s_i A_{ji} e_j\right)$$ (einskim but we have a single story house)

$$= \underbrace{-ds_i \wedge (s_i A_{ji} e_i)}_{\nabla \text{ of 1st term}} + \underbrace{\nabla(s_i A_{ji}) e_j \wedge ds_i e_i}_{\nabla \text{ of 2nd term}}$$

$$= \underbrace{-ds_i \wedge (s_i A_{ji} e_i)}_{\nabla \text{ of 1st term}} + (ds_i \wedge A_{ji} + s_i dA_{ji}) e_j - (s_i A_{ji}) \wedge (A_{kj} e_k)$$

$$= s_i \, dA_{ji} e_j + s_i A_{jl} A_{li} e_j$$

$$= dA + A \wedge A. \square$$

The upshot here is that curvature is the obstruction for the differential on the $\Omega^i(E)$ complex to square to 0.

We'll now consider the bundle $End(E)$. Given a connection $\nabla = d + A$ on E, we can define a connection on $End(E)$ using ∇.

We do so locally

Def A connection D_A on $\mathrm{End}(E)$ is locally of the form

$$D_A B := dB + AB - BA$$

where A is a matrix of 1-forms and B is a section of $\mathrm{End}(E)$, i.e., a matrix valued function.

This definition can be motivated in two ways.
First, connections on E are identified with matrices, which act on vectors by multiplication and other matrices via the lie bracket. The dB term is the vector part, i.e. the usual entry-wise derivative, and the $AB-BA$ term records the linear action of A on B.

The second way to motivate this is by decomposing $\mathrm{End}(E)$ as $\mathrm{End}(E) = E \otimes E^*$, and checking that tensoring with the connection on E^* obtained by dualizing gives the same thing.

Now that we have a connection on $\mathrm{End}(E)$, we again get a map $D_A : \Omega^0(\mathrm{End}\,E) \to \Omega^1(\mathrm{End}\,E)$, which we can also extend to a map $D_A : \Omega^i(\mathrm{End}\,E) \to \Omega^{i+1}(\mathrm{End}\,E)$.

Exercise Use the Leibniz rule to extend D_A to a map
$D_A : \Omega^i(\mathrm{End}\,E) \to \Omega^{i+1}(\mathrm{End}\,E)$.

Further, if A is a $SU(2)$ connection, B is $SU(2)$-valued, so $[A,B]$ also $SU(2)$-valued.
Therefore we can restrict $\mathrm{End}(E)$ to $\mathrm{ad}(E)$ and get the complex

$$\Omega^0(\mathrm{ad}\,E) \xrightarrow{D_A} \Omega^1(\mathrm{ad}\,E) \to \dots \quad (*)$$

lemma For $B \in \Omega^1(\mathrm{ad}\, E)$, $D_A B$ as defined above is the $D_A B$ defined

!! Sign issue?

Pf First, any $B \in \Omega^1(\mathrm{ad}E)$ can be expressed as a linear combination of 1-forms ω with coefficients constant $SU(2)$ matrices C. Thus we can assume $B = \omega C$.

Then $D_A B = D_A(\omega C)$

$$\left(\begin{array}{c}\text{Leibniz} \\ \text{and } \omega \text{ is a 1-form}\end{array}\right) = d\omega C - \omega \wedge \nabla C$$

$$\left(\text{Since } C \text{ const}\right) = d\omega C - \omega \wedge (dC + AC - CA)$$

$$\left(\begin{array}{l}\text{Since } \omega, A \text{ both} \\ 1\text{-forms}, \\ \omega \wedge A = -A \wedge \omega\end{array}\right) = d\omega C - \omega \wedge [A, C]$$

$$= d\omega C - (\underbrace{\omega \wedge A}) C + \omega C \wedge A$$

$$= d\omega C - (-A \wedge \omega) C + \omega C \wedge A$$

$$= d(\omega C) + A \wedge (\omega C) + (\omega C) \wedge A$$

$$= dB + A \wedge B + B \wedge A. \quad \square$$

Note that this proof, applied to $\omega = i$-form instead, shows in general that the map

$D_A B: \Omega^i(\mathrm{End}\, E) \to \Omega^{i+1}(\mathrm{End}\, E)$ is

$D_A B = dB + A \wedge B - B \wedge A$, for i even,

and $D_A B = dB + A \wedge B + B \wedge A$, for i odd.

Now, seemingly from nowhere, we'll define the Bianci Identity - this will be incredibly useful later.

Thm [Bianchi Identity]

If A is fixed connection, $D_A F_A = 0$

$F_A \in \Omega^2(\mathrm{ad}(E))$

Pf Since $F_A = dA + A \wedge A$,

$$D_A F_A = dF_A + A \wedge F_A - F_A \wedge A$$

$$= d(dA + A\wedge A) + A \wedge (dA + A\wedge A) - (dA + A\wedge A)\wedge A$$

$$= (d^2 A + d(A\wedge A)) + (A\wedge dA + A\wedge A\wedge A)) - (dA\wedge A + A\wedge A\wedge A)$$

$$= d^2 A + d(A\wedge A) + A\wedge dA + A\wedge A\wedge A) - dA\wedge A - A\wedge A\wedge A$$

(Since $d^2 A = 0$ and $dA\wedge A = -A\wedge dA$) $= (dA\wedge A + A\wedge dA) + A\wedge dA - dA\wedge A$

Sign issue?

$$= d^2 A + dA\wedge A + A\wedge dA + A\wedge dA + A\wedge A\wedge A - dA\wedge A - A\wedge A\wedge A$$

$$= 0$$

Now, let $D_A^* = *\, D_A *$. This gives us a codifferential

$$D_A^* : \Omega^2(\mathrm{ad}\, E) \to \Omega^1(\mathrm{ad}\, E)$$

We are now ready to return to discussing the critical points of the Yang-Mills Functional.

Yang-Mills Connections

We have seen that A is a critical point of YM IFF

$$\langle D_A B, F_A \rangle_{L^2} = 0 \quad \text{for all } B \in \Omega^1(\mathrm{ad}\, E)$$

$$\Leftrightarrow \quad \langle B, D_A^* F_A \rangle_{L^2} = 0$$

$$\Leftrightarrow \quad D_A^* F_A = 0$$

Def A connection such that $D_A^* F_A = 0$ is called a Yang Mills connection. These connections are the critical points of the YM functional.

Note the similarity with Hodge theory!

Last week, we saw that the cp of L^2 norm on forms are precisely the harmonic forms: α for which $d\alpha = d^*\alpha = 0$

If a connection A is Yang-Mills, then $D_A^* F_A = 0$. And by the Bianchi identity, we always have that $D_A F_A = 0$.

Therefore YM connections can be thought of as harmonic forms; the slogan here is that "YM theory is the $SU(2)$ version of Hodge theory."

The one caveat is that in general, $D_A * F_A$ is difficult to understand. In Hodge theory, d always exists, but here, both D^*_A and F_A depend on A.

It turns out that some YM connections satisfy a simpler condition, which also guarantees that they are global minima/maxima.

Lemma 1) if A is SD or ASD, then $D_A^* F_A = 0$

2) In fact, (A)SD are the global minima/maxima of the YM functional

Pf 1) This is immediate from Bianchi:

Since $D_A F_A = 0$ and $D_A^* = * D_A *$,

if $F_A = \pm * F_A$, then

$$\begin{aligned} D_A^* F_A &= * D_A * F_A \\ &= * D_A \pm F_A \\ &= \pm * D_A F_A \\ &= 0. \end{aligned}$$

2) For $F_A \in \Omega^2(\mathrm{ad}E) = \Omega^2_+(\mathrm{ad}E) \oplus \Omega^2_-(\mathrm{ad}E)$, it splits as $F_A = F_A^+ + F_A^-$, where $\langle F_A^+, F_A^- \rangle_{L^2} = 0$.

Thus $YM(\sigma) = \langle F_A, F_A \rangle_{L^2}$

$$= -\int_M \mathrm{tr}(F_A \wedge *F_A)$$

$$= \langle F_A^+ + F_A^-, F_A^+ + F_A^- \rangle_{L^2}$$

$$= \langle F_A^+, F_A^+ \rangle_{L^2} + \langle F_A^-, F_A^- \rangle_{L^2} + \langle F_A^+, F_A^- \rangle_{L^2} + \langle F_A^-, F_A^+ \rangle_{L^2}$$

$$= \|F_A^+\|_{L^2} + \|F_A^-\|_{L^2}$$

Now, recall the instanton number k defined as

$$k = -\langle c_2, [M] \rangle = \frac{-1}{8\pi^2} \int_M \mathrm{tr}(F_A \wedge F_A).$$

Since $F_A = *(*F_A)$,

$$k = \frac{1}{8\pi^2} \langle F_A, *F_A \rangle_{L^2}$$

$$= \frac{1}{8\pi^2} \langle F_A^+ + F_A^-, *(F_A^+ + F_A^-) \rangle_{L^2}$$

$$= \frac{1}{8\pi^2} \langle F_A^+ + F_A^-, F_A^+ - F_A^- \rangle_{L^2}$$

$$= \frac{1}{8\pi^2} \left(\langle F_A^+, F_A^+ \rangle_{L^2} + \langle F_A^+, -F_A^- \rangle_{L^2} + \langle F_A^-, F_A^+ \rangle_{L^2} + \langle F_A^-, -F_A^- \rangle \right)$$

$$= \frac{1}{8\pi^2} \left(\|F_A^+\|_{L^2} - \|F_A^-\|_{L^2} \right).$$

Therefore $8\pi^2 k = \|F_A^+\|_{L^2} - \|F_A^-\|_{L^2}$.

Combining this with $YM(\sigma) = \|F_A^+\|_{L^2} + \|F_A^-\|_{L^2}$, we see that

If $k>0$, then $YM \geq 8\pi^2 k$, with equality when $F_A^- \equiv 0$, ie, when A is self-dual. On the other hand, if if $k<0$, then $YM \geq 8\pi^2(-k)$ and equality is realized when $F_A^+ \equiv 0$, so A is anti-self-dual. $\square$